

MIMIC-III vs MIMIC-IV: Dataset Comparison Summary

Generated from: `mimic_analysis/compute_part1.py` (Part 1) · `mimic_analysis/build_patient_table2.py` (Part 2) **Pipeline:** `build_mimic3_wide.py` · `build_mimic4_wide.py`

Part 1: Wide-Table Overview (All ICU Stays)

Inclusion: All ICU stays; one row per stay. All statistics computed over `hr ≥ 0` rows only.

Definitions

Time axis. Both datasets extend to pre-ICU hours (negative `hr`):

	MIMIC-III	MIMIC-IV
Pre-ICU window	<code>hr = -12</code> to discharge	<code>hr = -24</code> to discharge
Reason for difference	Official <code>pivoted_lab.sql</code> uses a ±12 h fuzzy window; labs up to 12 h before admission map to <code>hr = 0</code> , so extending beyond -12 produces empty rows	<code>labevents</code> is queried by raw <code>charttime</code> with no restriction, enabling capture of labs up to 24 h before ICU admission

For `hr < 0`, ICU-sourced variables (vitals, GCS, vasopressors, urine output, ventilation, SOFA, SepsisLabel) are NULL by construction; only hospital laboratory values may be non-null.

SOFA score. Computed hourly for `hr ≥ 0`. Each of the six components (respiration, coagulation, liver, cardiovascular, CNS, renal) is scored from raw values at the current hour (no carry-forward), then the maximum over the preceding 24 hours is taken (`MAX(score) OVER ROWS BETWEEN 24 PRECEDING AND CURRENT ROW`). `sofa_24hours` = sum of the six 24 h-max component scores.

Sepsis label.

	MIMIC-III	MIMIC-IV
Suspected infection	Antibiotic + culture within [-24 h, +12 h]	Antibiotic + culture within [-48 h, +24 h]
Sepsis criterion	SOFA rise ≥ 2 from 24 h rolling minimum (<code>sofa_delta_24h ≥ 2</code>)	Absolute SOFA ≥ 2 (<code>sofa_24hours ≥ 2</code>)
Reference	Seymour et al. 2016 / MIMIC-III Sepsis Challenge	Singer et al. 2016 / <code>mimic-code sepsis3.sql</code>

`SepsisLabel = 1` from `onset_hr` onward within the stay; = 0 before onset; = NULL for `hr < 0`. A stay is classified as a **sepsis stay** if `SepsisLabel = 1` at any `hr ≥ 0`.

ICU LOS: $\text{MAX}(\text{hr})$ per stay (hours from admission to discharge). **Coverage rate:** proportion of stays with ≥ 1 non-null value for the variable at $\text{hr} = 0$.

Table 1. Dataset Characteristics

	MIMIC-III	MIMIC-IV
Scale		
Total ICU stays	52,840	94,444
Unique subjects	38,484	65,355
Total patient-hours ($\text{hr} \geq 0$)	5,274,012	8,322,254
ICU LOS, mean (h)	98.8	87.1
ICU LOS, median [IQR] (h)	51.0 [28.0, 100.0]	47.0 [26.0, 93.0]
LOS < 24 h, %	16.0%	19.9%
Hospital mortality, n (%)	6,516 (12.3%)	11,343 (12.0%)
Sepsis		
Sepsis stays, n (%)	19,809 (37.5%)	40,456 (42.8%)
Non-sepsis stays, n (%)	33,031 (62.5%)	53,988 (57.2%)
Sepsis subjects, n (%)	15,785 (41.0%)	31,390 (48.0%)
Sepsis onset, median [IQR] (h)	0 [0, 2]	2 [1, 6]
SOFA		
Max SOFA per stay, mean	4.6	4.7
Max SOFA per stay, median [IQR]	4 [2, 6]	4 [2, 7]
Max SOFA ≥ 2 , % of stays	81.4%	80.4%
Max SOFA ≥ 6 , % of stays	32.4%	33.8%
SOFA at sepsis onset, median [IQR]	0 [0, 2]	3 [2, 4]
SOFA Δ at sepsis onset (III only), median [IQR]	0 [0, 1]	—

Table 2. First ICU Unit by Volume

ICU Unit	MIMIC-III	MIMIC-IV
MICU	20,837 (39.4%)	20,699 (21.9%)
MICU/SICU	—	15,447 (16.4%)
CVICU	—	14,769 (15.6%)
CSRU	9,228 (17.5%)	—

ICU Unit	MIMIC-III	MIMIC-IV
SICU	8,794 (16.6%)	13,008 (13.8%)
CCU	7,639 (14.5%)	10,771 (11.4%)
TSICU	6,342 (12.0%)	10,474 (11.1%)
Neuro Intermediate	—	5,776 (6.1%)
Neuro SICU	—	1,750 (1.9%)

Table 3. Clinical Data Coverage at hr = 0 (% of stays with ≥ 1 non-null value)

Variable	MIMIC-III	MIMIC-IV
Heart rate	41.1%	55.8%
Systolic BP	39.7%	52.6%
Temperature	N/A †	38.7%
SpO ₂	39.5%	54.3%
GCS total	23.0%	26.8%
Creatinine	12.6%	10.4%
Platelet	14.2%	11.4%
PaO ₂	19.6%	16.0%
Bilirubin (total)	4.8%	5.0%
FiO ₂	4.7%	3.0%

† Temperature is not available as a standalone vital in the MIMIC-III wide table (arterial blood gas temperature only).

Key Observations

1. Sepsis rate: MIMIC-IV (42.8%) > MIMIC-III (37.5%)

The difference is driven by the **labelling criterion**, not a genuinely higher prevalence of sepsis.

MIMIC-III applies a delta criterion (SOFA rise ≥ 2 from a 24 h rolling minimum), which does not label patients who arrive critically ill and do not deteriorate further. As a result, the median SOFA at sepsis onset is **0 [0, 2]**, indicating that most flagged patients experienced a rise from a low baseline rather than an acute decompensation. MIMIC-IV applies an absolute criterion (SOFA ≥ 2 at time of suspected infection), capturing patients admitted with an already-elevated SOFA; their median onset SOFA is **3 [2, 4]**.

The clinical severity of the labelled cohorts is nearly identical across datasets (first-day SOFA median 5.0 [3.0, 7.0]; 30-day mortality 20.3% vs 20.6%), confirming that the additional labels in MIMIC-IV represent patients

missed by the structural blind spot of the delta criterion — those who arrive critically ill but do not deteriorate further — rather than a milder case mix.

2. Vitals coverage: MIMIC-IV (~55%) > MIMIC-III (~41%) at hr = 0

The gap reflects a documentation system transition: MIMIC-III was recorded primarily through CareVue (manual nursing entry), whereas MIMIC-IV transitioned to MetaVision (automated bedside monitor downloads), which captures vitals at higher frequency and completeness.

3. SOFA means aligned: MIMIC-III (4.6) ≈ MIMIC-IV (4.7)

Prior to correction, the MIMIC-III mean max SOFA was 8.3. The root cause was a structural defect in the renal SOFA urine-output logic, described in detail below.

In-Depth Analysis: Systematic Bias in Early Renal SOFA Scoring

Core Conclusion

When extracting the Renal SOFA score for the first 48 hours of an ICU admission, there is a massive gap between MIMIC-III and MIMIC-IV. Tracing the underlying SQL confirms: this is not a local data extraction bug, but a **severe boundary truncation defect in MIMIC-III's official script when handling cumulative metrics (urine output)**. The defect was structurally resolved in MIMIC-IV through an observation-time-aware mechanism.

I. Empirical Evidence: Systematic False Positives in MIMIC-III

ICU Hours Elapsed	MIMIC-III (Score = 4)	MIMIC-IV (Score = 4)	Observation
hr = 0	5.4%	0.5%	IV reflects true baseline of severe renal failure.
hr = 6	44.6% (peak)	—	III false positives peak; nearly half of patients deemed anuric.
hr = 24	49.5%	5.2%	IV completes first full 24 h window; ratio is stable.
hr = 48	8.6% (sudden drop)	7.5%	III exits the flawed rolling window; drops back to normal.

The absolute driver: urine output (UO). In MIMIC-III, 84.1% of abnormally high renal scores (≥ 2) are driven by UO alone:

- Pure UO trigger: **62.9%** — creatinine is normal; the patient is penalized solely by the UO logic.
- Pure creatinine trigger: 19.9%.

II. Defect Origin: Flaw in MIMIC-III's Official Script

The root cause lies in `pivoted_sofa.sql`. A fixed row-based window is applied blindly, producing two compounding errors: **early boundary truncation** and **extreme-value propagation**.

For snapshot metrics (respiration, coagulation), taking the min or max over 24 hours is unaffected by how long the patient has been in the ICU. Urine output is a **cumulative metric** requiring `SUM` — truncated sums produce systematically wrong totals.

MIMIC-III official `pivoted_sofa.sql`:

```
-- Root cause: mechanical 24-row backward rolling window
SUM(UrineOutput) OVER (
  PARTITION BY icustay_id
  ORDER BY hr
  ROWS BETWEEN 23 PRECEDING AND CURRENT ROW -- sums whatever rows exist,
  regardless of actual time span
) AS uo_24hr
```

Error chain:

1. **Truncated comparison** — At hr = 6 the engine finds only 7 rows. Normal urine over 7 hours (~400 mL) is compared directly against the 24 h clinical threshold (< 500 mL = score 3).
2. **False-positive explosion** — Seven hours of urine can rarely meet a 24 h passing mark; healthy kidneys are mass-classified as severely oliguric in the early hours.
3. **Ghost propagation** — The final SOFA uses `MAX(sofa_renal) OVER 24 PRECEDING`. A spuriously generated score of 4 at hr = 7 propagates forward for 24 hours, exiting the window completely only at hr = 48 — explaining the abrupt drop above.

III. The Fix: Observation-Time-Aware Logic in MIMIC-IV

MIMIC-IV's `sofa.sql` introduces `uo_tm_24hr` (accumulated observation duration) and refuses to score until the window is sufficiently covered.

MIMIC-IV official `sofa.sql`:

```
MAX(
  CASE
    -- Refuse to evaluate if observation window < 22 h
    WHEN uo.uo_tm_24hr >= 22 AND uo.uo_tm_24hr <= 30
    -- Normalise to a standard 24 h equivalent
    THEN uo.urineoutput_24hr / uo.uo_tm_24hr * 24
  END
) AS uo_24hr
```

During the first ~22 hours, `uo_24hr` is forcibly NULL — no UO-driven score is possible. The scoring engine falls back entirely to admission serum creatinine, which is why MIMIC-IV shows a realistic 0.5% severe-failure rate at hr = 0. This logic was ported into the MIMIC-III pipeline (`build_mimic3_wide.py`, `step07_uo`), after which both datasets align: max SOFA mean 4.6 (III) vs 4.7 (IV).

Part 2: Patient-Level Cohort Summary

Inclusion: One row per patient, first ICU stay only.

Definitions:

- **SOFA:** first-day maximum SOFA score
- **SIRS:** first-day maximum SIRS score
- **Comorbidity index:** Elixhauser van Walraven score (MIMIC-III, official mimic-code ICD-9 implementation) / Charlson Comorbidity Index with age adjustment (MIMIC-IV, official mimic-code ICD-9 + ICD-10 implementation). *Scores are not directly comparable across datasets.*
- **Mechanical ventilation:** any invasive/non-invasive ventilation or tracheostomy during the first ICU stay
- **30-day mortality:** measured from first ICU admission time
- **Race/ethnicity:** grouped into 7 categories (White, Black, Hispanic, Asian, Native, Other, Unknown)

Table 4. MIMIC-III Patient Characteristics

Variable	All patients (N = 38,484)	Survivors (N = 34,070, 88.5%)	Non-survivors (N = 4,414, 11.5%)	P (survival)	Sepsis (N = 13,436, 34.9%)	Non-sepsis (N = 25,048, 65.1%)	P (sepsis)
Age (y), median [IQR]	66.0 [52.0, 78.0]	65.0 [52.0, 77.0]	74.0 [60.0, 83.0]	<0.001	68.0 [55.0, 79.0]	64.0 [51.0, 77.0]	<0.001
Male, n (%)	21,793 (56.6%)	19,461 (57.1%)	2,332 (52.8%)	<0.001	7,726 (57.5%)	14,067 (56.2%)	0.011
BMI (kg/m ²), mean ± SD	28.4 ± 7.0	28.5 ± 7.0	27.5 ± 7.1	<0.001	28.8 ± 7.4	28.3 ± 6.8	<0.001
Race, n (%)				<0.001			<0.001
White	27,457 (71.3%)	24,451 (71.8%)	3,006 (68.1%)		9,783 (72.8%)	17,674 (70.6%)	
Black	2,949 (7.7%)	2,695 (7.9%)	254 (5.8%)		1,081 (8.0%)	1,868 (7.5%)	
Hispanic	1,252 (3.3%)	1,166 (3.4%)	86 (1.9%)		421 (3.1%)	831 (3.3%)	
Asian	909 (2.4%)	806 (2.4%)	103 (2.3%)		350 (2.6%)	559 (2.2%)	
Native American	20 (0.1%)	19 (0.1%)	1 (0.0%)		8 (0.1%)	12 (0.0%)	

Variable	All patients (N = 38,484)	Survivors (N = 34,070, 88.5%)	Non-survivors (N = 4,414, 11.5%)	P (survival)	Sepsis (N = 13,436, 34.9%)	Non-sepsis (N = 25,048, 65.1%)	P (sepsis)
Other	1,041 (2.7%)	935 (2.7%)	106 (2.4%)		389 (2.9%)	652 (2.6%)	
Unknown	4,856 (12.6%)	3,998 (11.7%)	858 (19.4%)		1,404 (10.4%)	3,452 (13.8%)	
Comorbidity index †, median [IQR]	5.0 [0.0, 12.0]	5.0 [0.0, 11.0]	11.0 [5.0, 17.0]	<0.001	9.0 [3.0, 15.0]	4.0 [0.0, 10.0]	<0.001
SIRS, first-day max, median [IQR]	3.0 [2.0, 3.0]	3.0 [2.0, 3.0]	3.0 [2.0, 4.0]	<0.001	3.0 [2.0, 4.0]	2.0 [2.0, 3.0]	<0.001
SOFA, first-day, median [IQR]	3.0 [1.0, 5.0]	3.0 [1.0, 5.0]	5.0 [3.0, 9.0]	<0.001	5.0 [3.0, 7.0]	2.0 [1.0, 4.0]	<0.001
ICU LOS (d), median [IQR]	2.0 [1.0, 4.0]	2.0 [1.0, 4.0]	3.0 [1.0, 7.0]	<0.001	3.0 [2.0, 8.0]	2.0 [1.0, 3.0]	<0.001
Mechanical ventilation, n (%)	19,483 (50.6%)	16,271 (47.8%)	3,212 (72.8%)	<0.001	8,749 (65.1%)	10,734 (42.9%)	<0.001
30-day mortality, n (%)	5,234 (13.6%)	1,073 (3.1%)	4,161 (94.3%)	<0.001	2,730 (20.3%)	2,504 (10.0%)	<0.001
Hospital mortality, n (%)	4,414 (11.5%)	0 (0.0%)	4,414 (100.0%)	<0.001	2,385 (17.8%)	2,029 (8.1%)	<0.001

Table 5. MIMIC-IV Patient Characteristics

Variable	All patients (N = 65,355)	Survivors (N = 58,275, 89.2%)	Non-survivors (N = 7,080, 10.8%)	P (survival)	Sepsis (N = 27,326, 41.8%)	Non-sepsis (N = 38,029, 58.2%)	P (sepsis)
Age (y), median [IQR]	66.0 [54.0, 78.0]	66.0 [54.0, 77.0]	73.0 [61.0, 83.0]	<0.001	67.0 [56.0, 78.0]	65.0 [53.0, 77.0]	<0.001

Variable	All patients (N = 65,355)	Survivors (N = 58,275, 89.2%)	Non-survivors (N = 7,080, 10.8%)	P (survival)	Sepsis (N = 27,326, 41.8%)	Non-sepsis (N = 38,029, 58.2%)	P (sepsis)
Male, n (%)	36,714 (56.2%)	32,875 (56.4%)	3,839 (54.2%)	<0.001	15,851 (58.0%)	20,863 (54.9%)	<0.001
BMI (kg/m ²), mean ± SD	28.8 ± 7.4	28.9 ± 7.3	28.7 ± 8.3	0.125	29.1 ± 7.7	28.5 ± 7.1	<0.001
Race, n (%)				<0.001			
White	43,089 (65.9%)	38,990 (66.9%)	4,099 (57.9%)		17,890 (65.5%)	25,199 (66.3%)	
Black	6,016 (9.2%)	5,470 (9.4%)	546 (7.7%)		2,303 (8.4%)	3,713 (9.8%)	
Hispanic	2,351 (3.6%)	2,177 (3.7%)	174 (2.5%)		933 (3.4%)	1,418 (3.7%)	
Asian	1,980 (3.0%)	1,769 (3.0%)	211 (3.0%)		800 (2.9%)	1,180 (3.1%)	
Native American	127 (0.2%)	117 (0.2%)	10 (0.1%)		58 (0.2%)	69 (0.2%)	
Other	2,416 (3.7%)	2,183 (3.7%)	233 (3.3%)		987 (3.6%)	1,429 (3.8%)	
Unknown	9,376 (14.3%)	7,569 (13.0%)	1,807 (25.5%)		4,355 (15.9%)	5,021 (13.2%)	
Comorbidity index ‡, median [IQR]	4.0 [2.0, 6.0]	4.0 [2.0, 6.0]	6.0 [4.0, 8.0]	<0.001	5.0 [3.0, 7.0]	4.0 [2.0, 6.0]	<0.001
SIRS, first-day max, median [IQR]	2.0 [1.0, 2.0]	2.0 [1.0, 2.0]	2.0 [2.0, 3.0]	<0.001	2.0 [1.0, 3.0]	2.0 [1.0, 2.0]	<0.001
SOFA, first-day, median [IQR]	3.0 [1.0, 6.0]	3.0 [1.0, 5.0]	6.0 [3.0, 9.0]	<0.001	5.0 [3.0, 7.0]	2.0 [1.0, 4.0]	<0.001
ICU LOS (d), median [IQR]	1.9 [1.1, 3.7]	1.9 [1.1, 3.4]	2.7 [1.1, 6.4]	<0.001	3.0 [1.5, 6.4]	1.5 [0.9, 2.5]	<0.001
Mechanical ventilation, n (%)	28,339 (43.4%)	23,514 (40.4%)	4,825 (68.1%)	<0.001	17,242 (63.1%)	11,097 (29.2%)	<0.001

Variable	All patients (N = 65,355)	Survivors (N = 58,275, 89.2%)	Non-survivors (N = 7,080, 10.8%)	P (survival)	Sepsis (N = 27,326, 41.8%)	Non-sepsis (N = 38,029, 58.2%)	P (sepsis)
30-day mortality, n (%)	8,964 (13.7%)	2,208 (3.8%)	6,756 (95.4%)	<0.001	5,620 (20.6%)	3,344 (8.8%)	<0.001
Hospital mortality, n (%)	7,080 (10.8%)	0 (0.0%)	7,080 (100.0%)	<0.001	4,671 (17.1%)	2,409 (6.3%)	<0.001

† Elixhauser van Walraven score (MIMIC-III): 29-condition weighted index based on secondary ICD-9 diagnoses (Quan et al. 2005; van Walraven et al. 2009). Scores range from approximately −19 to +89; higher scores indicate greater predicted in-hospital mortality risk. Negative weights apply to certain conditions (e.g., drug abuse −7, obesity −4).

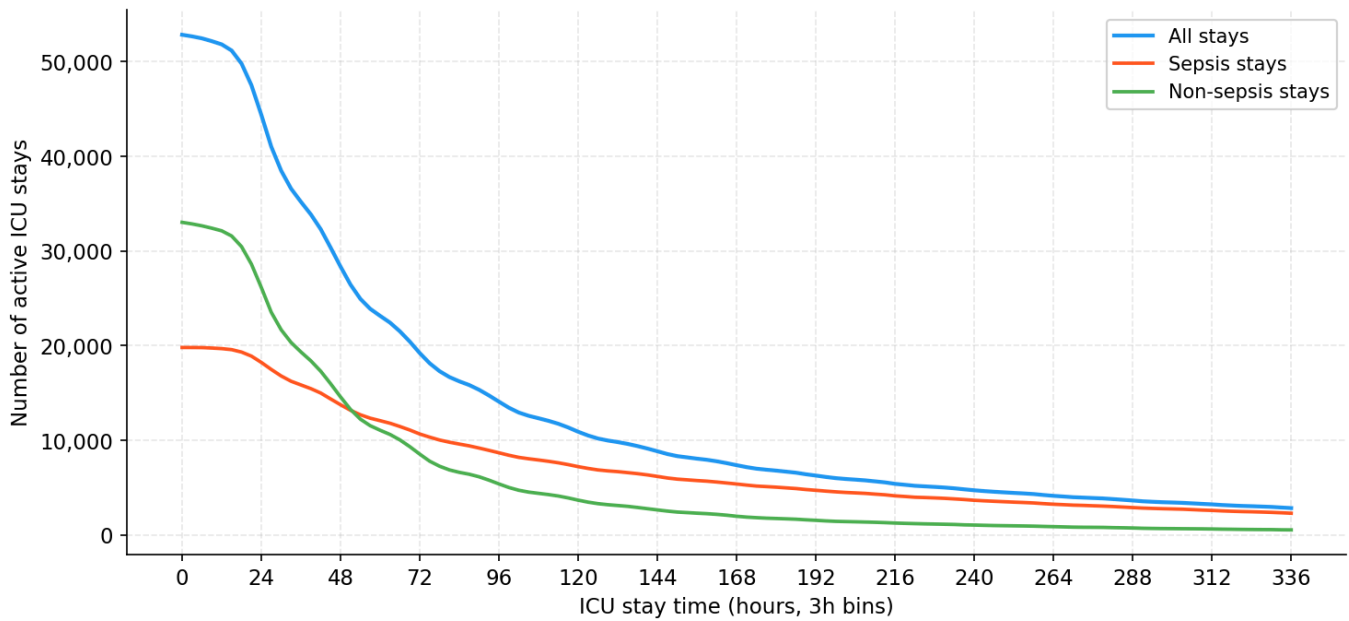
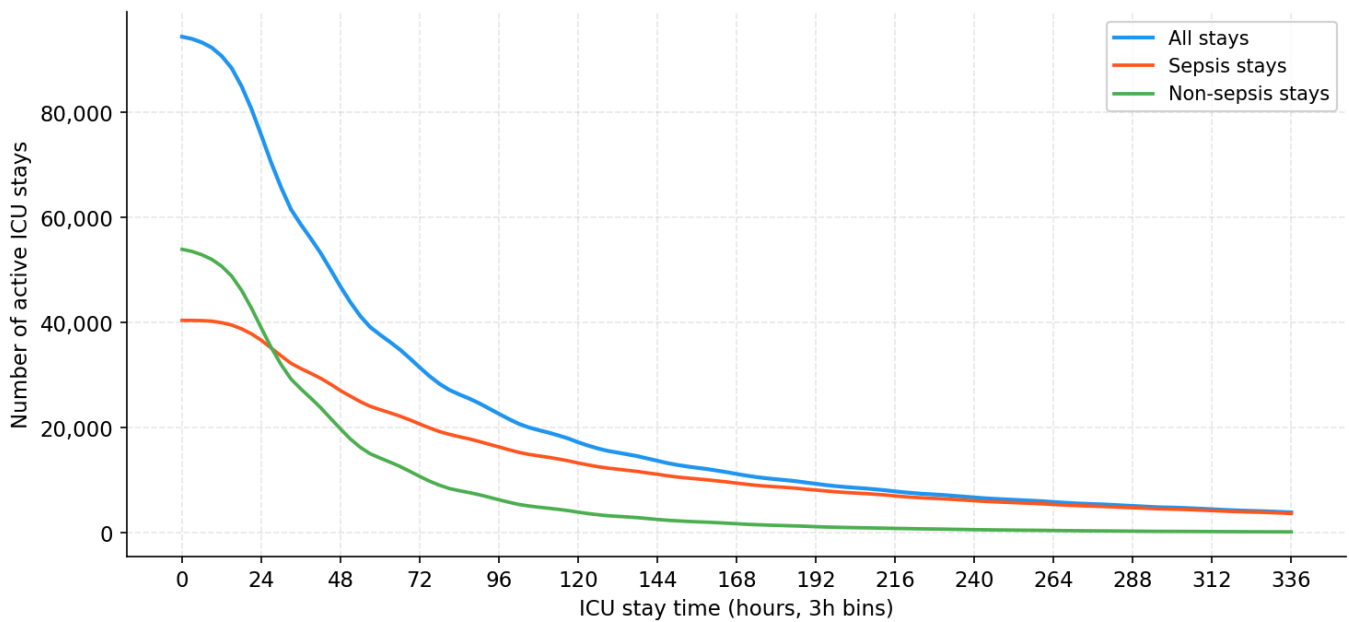
‡ Charlson Comorbidity Index with age adjustment (MIMIC-IV): 17-condition index based on ICD-9 and ICD-10 diagnoses (Charlson et al. 1987; Quan et al. 2011). Age contributes 0–4 points (1 point per decade above age 50). Scores range from 0 to approximately 30; higher scores indicate greater comorbidity burden. The two comorbidity indices are not directly comparable.

Part 3: Temporal Dynamics across ICU Stay

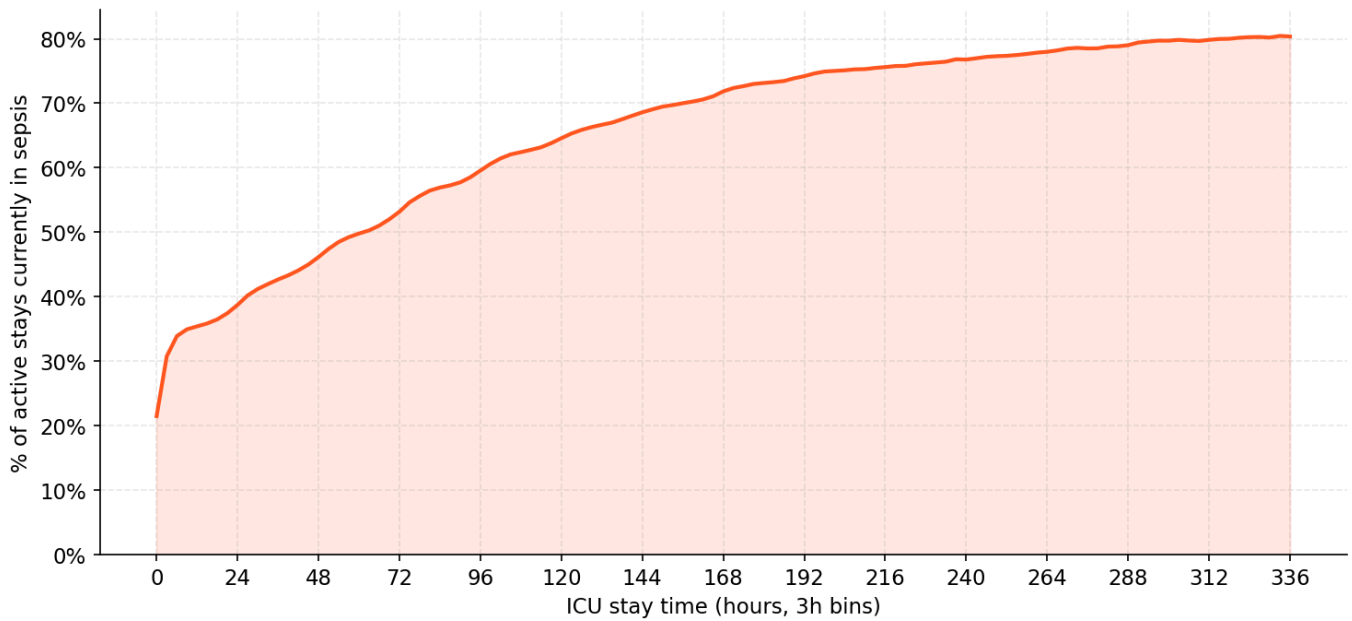
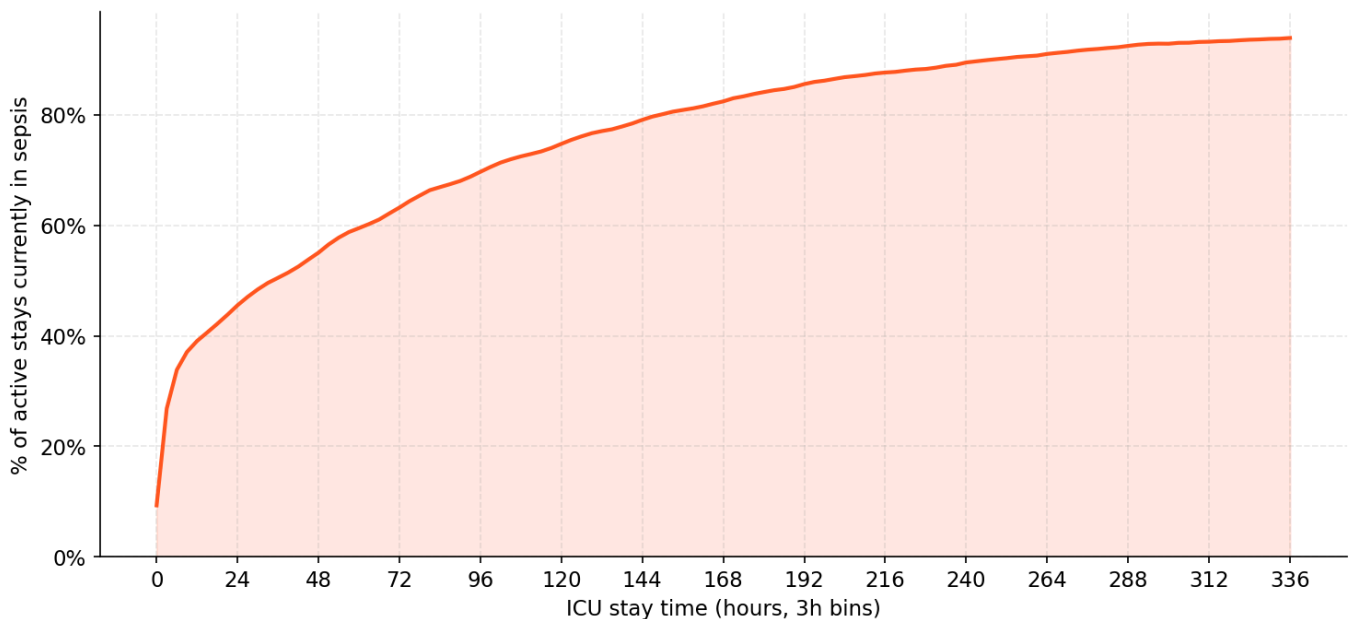
All figures use 3-hour time bins (x-axis: 0–336 h). Scripts: [mimic_analysis/plot_icu_curves.py](#).

Figure 1. Active ICU Stays over Time

Each curve shows the number of stays still ongoing at a given ICU hour — i.e., stays whose length of stay $\geq t$. Non-sepsis stays discharge much faster (green flattens near zero by ~72 h), while sepsis stays persist longer and come to dominate the late ICU population. MIMIC-IV has roughly 1.8 $\times$ more total stays than MIMIC-III at every time point, consistent with its larger overall cohort.

MIMIC-III**MIMIC-III — Active ICU Stays over Time****MIMIC-IV****MIMIC-IV — Active ICU Stays over Time****Figure 2. Sepsis Prevalence among Active Stays**

At each hour t , the curve shows the percentage of still-active stays whose sepsis onset has already occurred (i.e., $\text{onset_hr} \leq t$). The monotone increase reflects two compounding effects: new sepsis diagnoses accumulate over time, and non-sepsis patients discharge earlier, shrinking the denominator. By 336 h, over 80% (MIMIC-III) and 92% (MIMIC-IV) of remaining ICU stays have been labelled sepsis — a clear illustration of length-of-stay bias in long-stay cohorts.

MIMIC-III**MIMIC-III — Sepsis Prevalence among Active ICU Stays****MIMIC-IV****MIMIC-IV — Sepsis Prevalence among Active ICU Stays****Figure 3. Mean SOFA Score over ICU Stay**

Mean SOFA is computed at each 3-hour bin across all stays still active at that time, stratified by sepsis status (classified by the full-stay label). In both datasets the mean rises steeply during the first 24 hours as the 24-hour rolling maximum window fills, then plateaus. Sepsis stays maintain a consistently higher SOFA throughout (III: ~5, IV: ~5–6 vs non-sepsis ~3 and ~2.5 respectively), confirming a persistent severity gap between the two cohorts that does not close over the ICU stay.

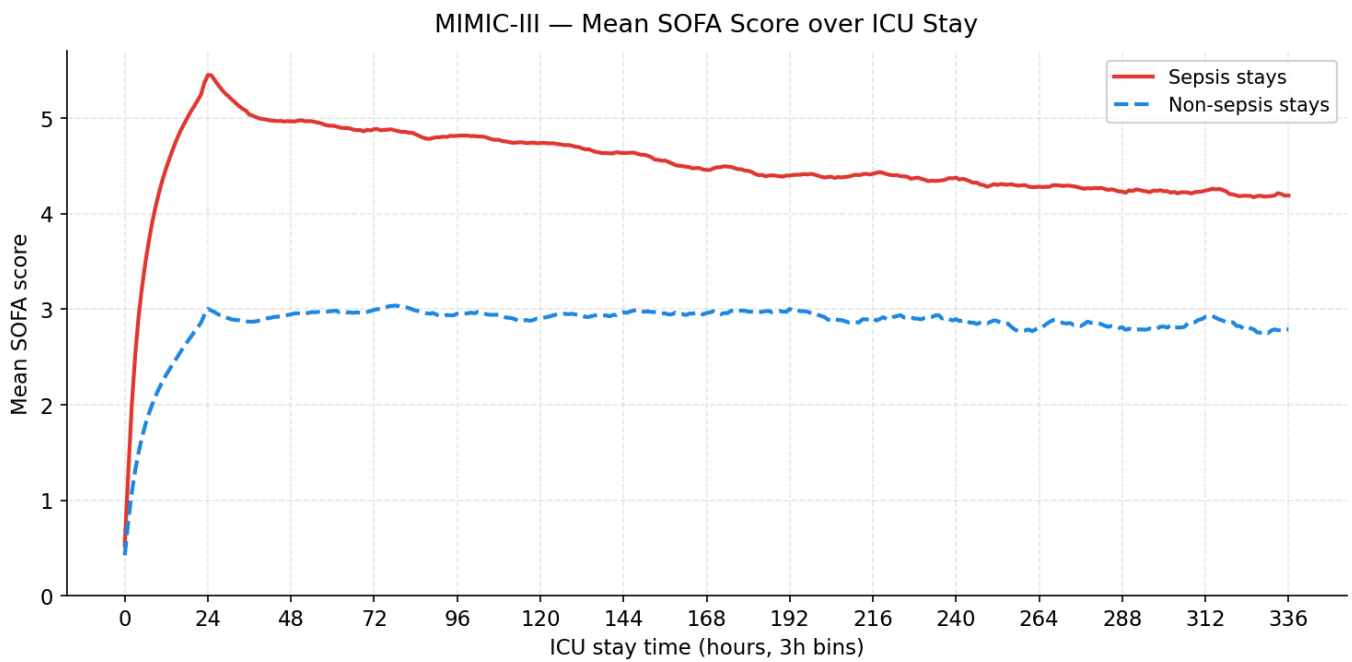
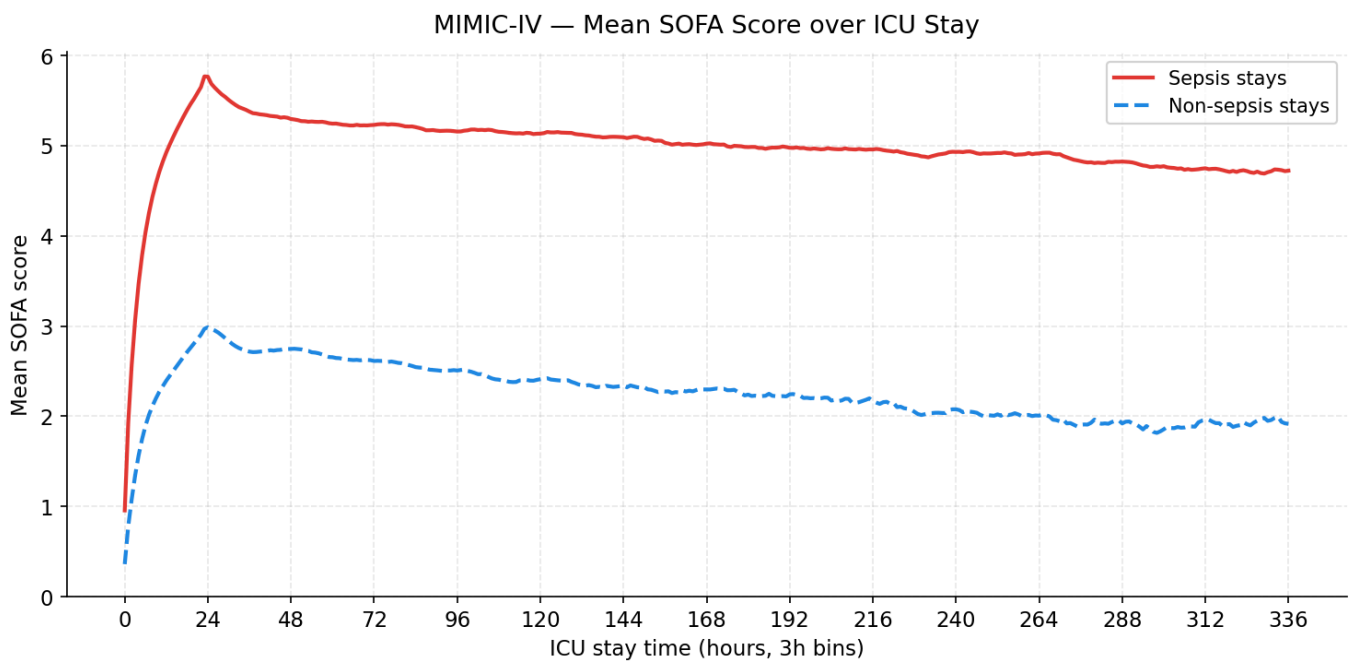
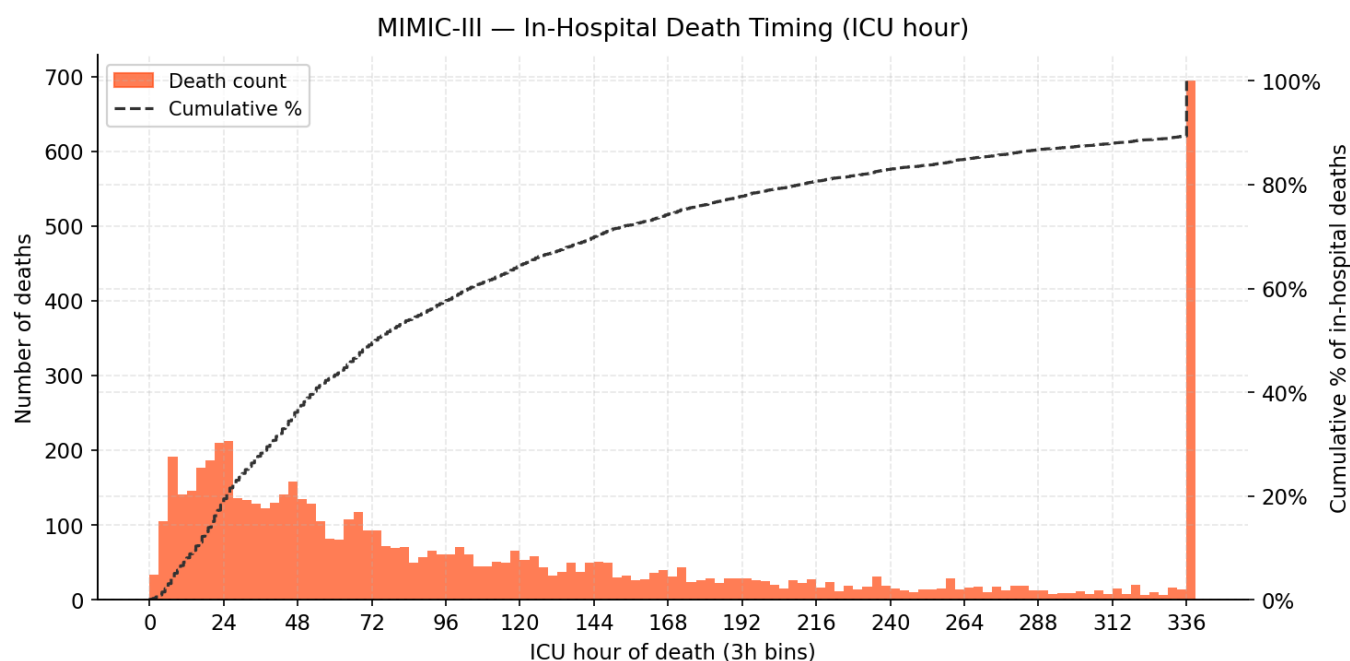
MIMIC-III**MIMIC-IV**

Figure 4. In-Hospital Death Timing

The histogram shows when in-hospital deaths occur relative to ICU admission, approximated as the last recorded ICU hour for each death stay. Deaths are concentrated in the first 48–72 hours, with the highest incidence around ICU hours 12–36. The dashed cumulative curve shows that ~50% of all in-hospital deaths occur within the first 48 h (MIMIC-III) or 72 h (MIMIC-IV). The spike at h = 336 is an artefact of clipping: stays longer than 14 days are capped at the plot boundary.

MIMIC-III



MIMIC-IV

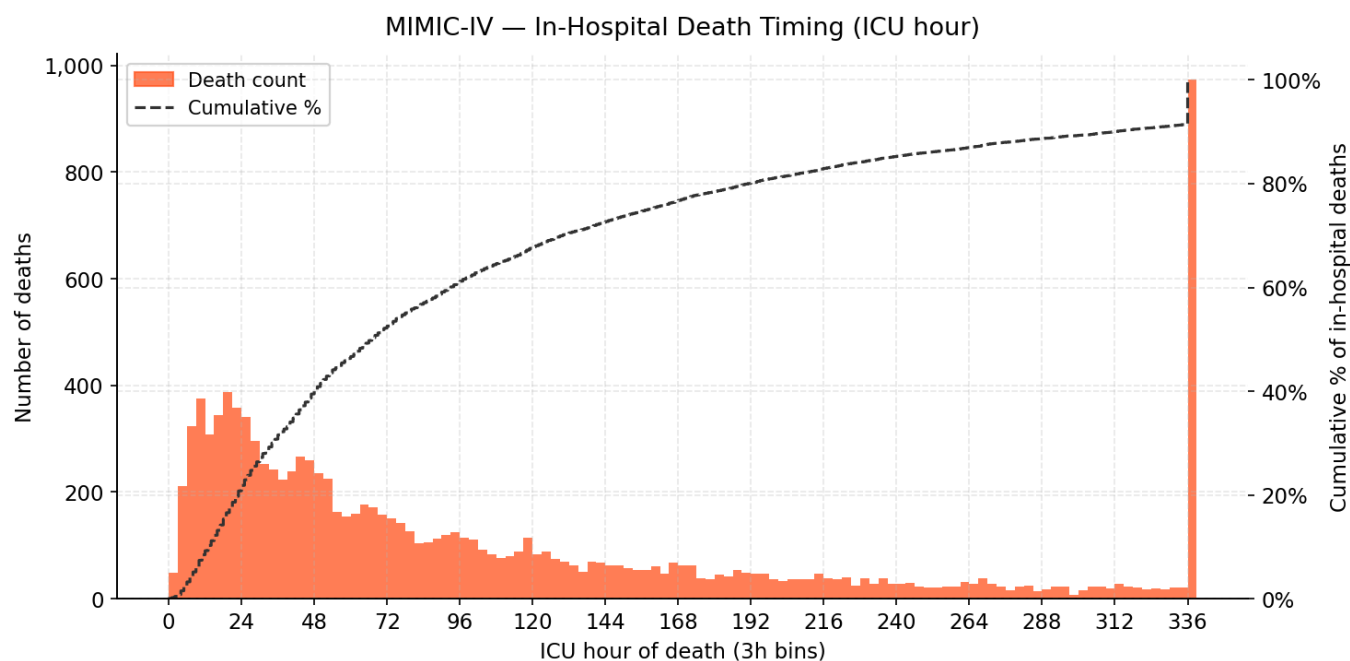
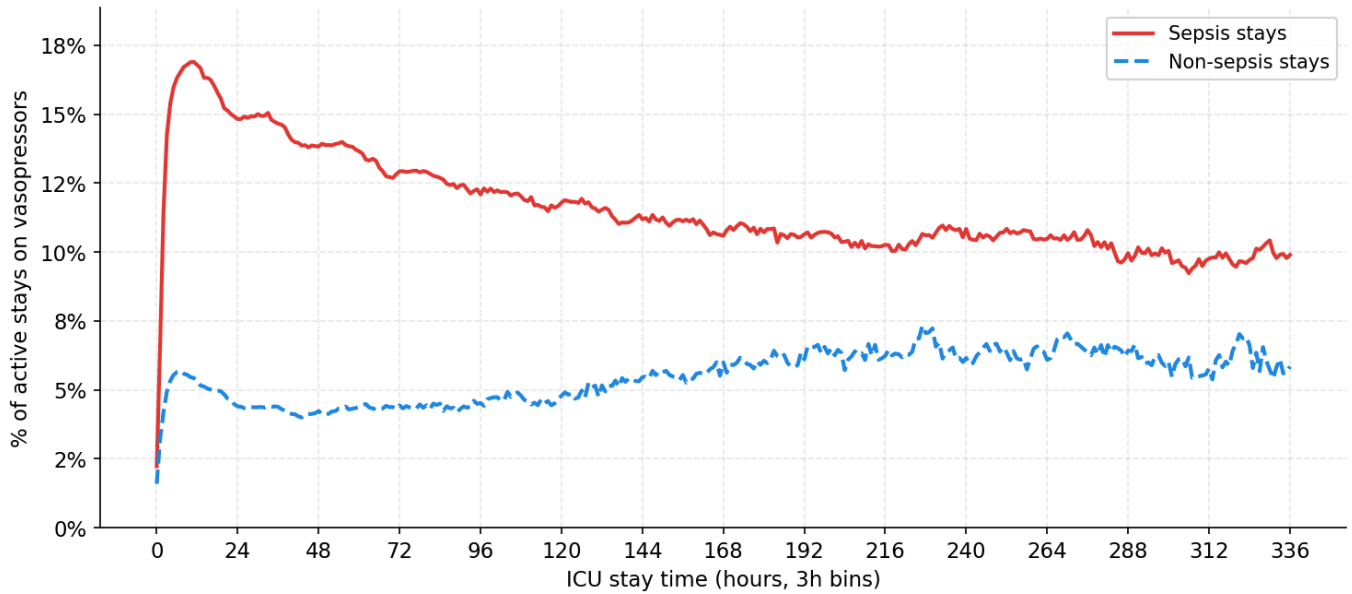


Figure 5. Vasopressor Use Rate over ICU Stay

The y-axis shows, at each 3-hour bin, the percentage of currently active stays receiving any vasopressor (norepinephrine, epinephrine, dopamine, or vasopressin). Both axes are on the same scale to allow direct comparison. In both datasets sepsis stays peak at ~17–18% around hours 6–12, then gradually decline as patients are weaned. Non-sepsis stays show much lower rates throughout (III: ~4–6%, IV: ~1–3%), reflecting the higher haemodynamic instability of septic shock. The late-stay non-sepsis increase is a selection effect: only the most critically ill non-sepsis patients remain in the ICU beyond day 10.

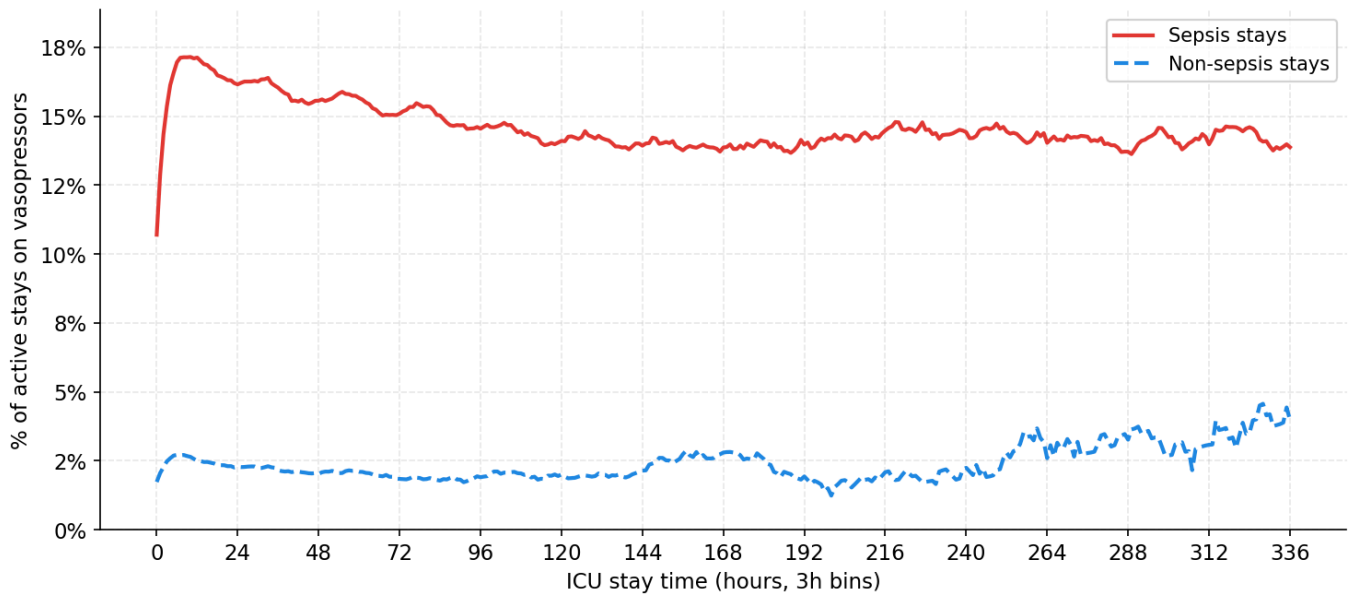
MIMIC-III

MIMIC-III — Vasopressor Use Rate over ICU Stay
(any of: norepinephrine / epinephrine / dopamine / vasopressin)



MIMIC-IV

MIMIC-IV — Vasopressor Use Rate over ICU Stay
(any of: norepinephrine / epinephrine / dopamine / vasopressin)



Part 4: Subject-Level ICU Stay Patterns

Unit of analysis: one row per unique subject (first vs. subsequent stays). Sepsis label per stay = 1 if `SepsisLabel = 1` at any `hr ≥ 0` within that stay, else 0.

5.1 Distribution of ICU Stays per Subject

MIMIC-III: **38,484** unique subjects — 8,300 (21.6%) have more than one ICU stay.

MIMIC-IV: **65,355** unique subjects — 16,241 (24.9%) have more than one ICU stay.

All non-sepsis = all stays for this subject are labelled 0; *All sepsis* = all stays labelled 1; *Mixed* = at least one 0 and at least one 1 across stays.

MIMIC-III

ICU stays	Subjects, n (%)	≤ n stays (%)	All non-sepsis	All sepsis	Mixed
1	30,184 (78.4%)	78.4%	19,943 (66.1%)	10,241 (33.9%)	0 (0.0%)
2	5,449 (14.2%)	92.6%	2,215 (40.6%)	1,057 (19.4%)	2,177 (40.0%)
3	1,557 (4.0%)	96.6%	404 (25.9%)	190 (12.2%)	963 (61.8%)
4	630 (1.6%)	98.3%	88 (14.0%)	64 (10.2%)	478 (75.9%)
5	290 (0.8%)	99.0%	34 (11.7%)	27 (9.3%)	229 (79.0%)
6	151 (0.4%)	99.4%	6 (4.0%)	4 (2.6%)	141 (93.4%)
7	82 (0.2%)	99.6%	6 (7.3%)	3 (3.7%)	73 (89.0%)
8	40 (0.1%)	99.7%	0 (0.0%)	2 (5.0%)	38 (95.0%)
9	22 (0.1%)	99.8%	1 (4.5%)	0 (0.0%)	21 (95.5%)
10	23 (0.1%)	99.9%	0 (0.0%)	0 (0.0%)	23 (100.0%)
≥11	56 (0.1%)	100.0%	2 (3.6%)	0 (0.0%)	54 (96.4%)

MIMIC-IV

ICU stays	Subjects, n (%)	≤ n stays (%)	All non-sepsis	All sepsis	Mixed
1	49,114 (75.1%)	75.1%	29,212 (59.5%)	19,902 (40.5%)	0 (0.0%)
2	10,341 (15.8%)	91.0%	3,825 (37.0%)	2,242 (21.7%)	4,274 (41.3%)
3	3,206 (4.9%)	95.9%	670 (20.9%)	455 (14.2%)	2,081 (64.9%)
4	1,289 (2.0%)	97.9%	169 (13.1%)	125 (9.7%)	995 (77.2%)
5	580 (0.9%)	98.7%	41 (7.1%)	40 (6.9%)	499 (86.0%)
6	308 (0.5%)	99.2%	22 (7.1%)	12 (3.9%)	274 (89.0%)
7	158 (0.2%)	99.5%	5 (3.2%)	3 (1.9%)	150 (94.9%)
8	110 (0.2%)	99.6%	7 (6.4%)	3 (2.7%)	100 (90.9%)
9	55 (0.1%)	99.7%	3 (5.5%)	2 (3.6%)	50 (90.9%)
10	53 (0.1%)	99.8%	2 (3.8%)	1 (1.9%)	50 (94.3%)
≥11	141 (0.2%)	100.0%	9 (6.4%)	1 (0.7%)	131 (92.9%)

5.2 Sepsis Sequence Patterns

Each subject's ICU stays are ordered by admission time and collapsed to a binary sepsis label per stay (0 = no sepsis, 1 = sepsis), then concatenated into a sequence such as 0->1->0. Proportion within group = share among all subjects with the same number of ICU stays.

MIMIC-III: **301** unique sequences in total (30 with ≤ 4 stays).

MIMIC-IV: **514** unique sequences in total (30 with ≤ 4 stays).

MIMIC-III

Sequences with 1–4 ICU Stays (all combinations)

n stays	Sequence	Subjects, n	% within group
1	0	19,943	66.1%
1	1	10,241	33.9%
2	0->0	2,215	40.6%
2	0->1	1,194	21.9%
2	1->1	1,057	19.4%
2	1->0	983	18.0%
3	0->0->0	404	25.9%
3	0->0->1	220	14.1%
3	1->1->1	190	12.2%
3	0->1->1	170	10.9%
3	1->0->1	158	10.1%
3	1->0->0	157	10.1%
3	0->1->0	148	9.5%
3	1->1->0	110	7.1%
4	0->0->0->0	88	14.0%
4	1->1->1->1	64	10.2%
4	0->0->0->1	55	8.7%
4	0->0->1->1	42	6.7%
4	1->0->1->1	42	6.7%
4	0->1->0->0	41	6.5%
4	1->0->0->1	37	5.9%
4	1->0->0->0	34	5.4%
4	0->1->1->1	34	5.4%

n stays	Sequence	Subjects, n	% within group
4	0->0->1->0	33	5.2%
4	0->1->0->1	31	4.9%
4	1->1->1->0	29	4.6%
4	0->1->1->0	28	4.4%
4	1->1->0->1	25	4.0%
4	1->0->1->0	24	3.8%
4	1->1->0->0	23	3.7%

Sequences with ≥5 ICU Stays (summary)

n stays	Subjects	Unique patterns	Most common sequence	Count
5	290	32	0->0->0->0->0	34
6	151	52	0->0->0->0->0->1	8
7	82	51	0->0->0->0->0->0->0	6
8	40	36	0->1->1->0->1->0->0->0	2
9	22	22	0->1->1->1->1->1->0->0->1	1
10	23	22	1->0->0->1->1->1->0->1->1->1	2
11	12	12	1->0->0->0->0->1->1->0->1->0->1	1
12	9	9	0->0->1->0->1->0->0->0->1->0->0->1	1
13	9	9	0->0->1->1->0->1->0->0->0->1->1->1->1	1
14	4	4	1->1->1->1->1->1->1->1->0->1->1->1->0	1
15	6	6	1->1->0->1->1->1->0->0->1->1->1->1->0->1->1	1
16	1	1	1->1->1->1->0->1->1->1->1->1->0->1->1->0->0->1	1
17	2	2	0->1->1->1->1->0->0->1->1->1->0->1->1->0->1->1->0	1
18	2	2	0->1->0->0->0->0->0->1->0->0->0->1->0->1->0->0->0->0	1
19	1	1	0->1->0->1->0->0->1->0->1->0->0->0->1->1->1->1->1->0->1	1
21	2	2	0->0->0->0->0->0->0->1->1->0->0->1->0->0->0->0->0->1->0->0->0	1

n stays	Subjects	Unique patterns	Most common sequence	Count
22	1	1	0->0->1->0->0->0->1->0->0->0->1->1->1->1->1->1->1->0->1->1->1->0->1	1
23	1	1	0->0->0->0->0->1->0->0->0->0->0->1->0->1->0->1->0->0->0->0->0->0->0	1
25	2	2	1->0->1->0->1->1->0->0->0->1->0->0->1->0->0->0->0->0->1->0->1->0->0->0->1->1	1
30	1	1	0->0	1
35	1	1	0->0->0->0->0->0->0->0->1->1->0->0->0->0->0->1->0->0->0->0->0->0->0->0->0->0->0->0->0->0->0	1
38	1	1	0->0->0->0->0->0->1->1->1->0->0->0->1->0->0->1->1->0->0->0->0->0->0->0->0->0->0->0->1->0->0->0->1->0->0->1->0->0->0	1
41	1	1	0->1->1->0->0->0->0->0->1->0->0->0->0->0->0->0->0->0->1->0->0->0->0->0->1->0->0->0->1->0->0->0->0	1

MIMIC-IV

Sequences with 1–4 ICU Stays (all combinations)

n stays	Sequence	Subjects, n	% within group
1	0	29,212	59.5%
1	1	19,902	40.5%
2	0->0	3,825	37.0%
2	1->1	2,242	21.7%
2	1->0	2,231	21.6%
2	0->1	2,043	19.8%
3	0->0->0	670	20.9%
3	1->1->1	455	14.2%
3	1->0->0	427	13.3%
3	0->0->1	365	11.4%
3	1->0->1	352	11.0%

n stays	Sequence	Subjects, n	% within group
3	1->1->0	335	10.4%
3	0->1->1	306	9.5%
3	0->1->0	296	9.2%
4	0->0->0->0	169	13.1%
4	1->1->1->1	125	9.7%
4	0->0->0->1	97	7.5%
4	1->0->1->1	88	6.8%
4	1->0->0->0	82	6.4%
4	0->1->0->0	80	6.2%
4	1->1->0->1	78	6.1%
4	1->0->0->1	77	6.0%
4	1->1->0->0	72	5.6%
4	0->0->1->1	71	5.5%
4	0->0->1->0	68	5.3%
4	0->1->1->1	66	5.1%
4	1->1->1->0	64	5.0%
4	0->1->0->1	55	4.3%
4	0->1->1->0	49	3.8%
4	1->0->1->0	48	3.7%

Sequences with ≥5 ICU Stays (summary)

n stays	Subjects	Unique patterns	Most common sequence	Count
5	580	32	0->0->0->0->0	41
6	308	61	0->0->0->0->0->0	22
7	158	81	1->0->1->1->1->1->1	5
8	110	75	0->0->0->0->0->0->0->0	7
9	55	47	1->1->1->0->0->1->1->1->1	3
10	53	51	0->0->0->0->0->0->0->0->0->0	2
11	43	40	0->0->0->0->0->0->0->0->0->0->0	4

n stays	Subjects	Unique patterns	Most common sequence	Count
12	20	20	1->1->0->0->0->0->1->1->1->1->0	1
13	17	17	1->1->0->1->0->1->0->1->1->1->1->1->0	1
14	6	6	1->1->0->0->0->1->0->0->1->0->0->0->0->0	1
15	10	10	1->0->0->0->0->1->1->1->1->0->1->0->1->0->1	1
16	9	9	1->0->1->1->0->1->1->1->1->1->0->1->0->1->1->1	1
17	4	4	1->0->0->0->0->1->0->1->0->0->1->1->0->0->1->1-> >1	1
18	11	10	0->0->0->0->0->0->0->0->0->0->0->0->0->0->0->0-> >0->0	2
19	2	2	0->0->0->0->1->0->0->1->1->1->1->1->1->0->0->0-> >0->0->1	1
20	3	3	1->0->0->0->1->0->1->0->0->0->1->1->1->0->1->1-> >0->1->0->0	1
22	3	3	1->1->0->1->1->1->1->1->0->1->0->1->1->1->0->1-> >0->1->1->0->0->1	1
24	3	3	1->1->1->0->1->0->1->1->1->1->1->1->0->1->0->0->0-> >0->0->0->0->1->1->1->1	1
25	3	3	1->0->1->1->1->1->1->1->1->1->1->1->0->1->1->0-> >1->1->1->0->0->1->1->1->1	1
26	1	1	0->0->0->0->0->1->0->0->0->0->0->0->0->0->0->0->0-> >0->0->0->0->0->0->0->0->0	1
27	1	1	0->0->0->0->0->0->0->1->0->0->0->0->0->0->0->0->0-> >0->1->1->0->0->1->0->0->1->0->1	1
30	1	1	0->0->0->0->0->0->0->0->0->0->0->0->0->0->0->1-> >0->0->0->0->0->1->0->1->1->0->1->1->0->1	1
31	1	1	1->1->0->0->0->1->0->0->0->1->0->0->0->1->1->1-> >1->0->0->0->1->0->0->0->0->0->0->0->1->0	1
34	1	1	0->0->0->0->0->0->0->0->0->1->0->0->0->0->0->0->0-> >0->0->0->0->0->0->0->0->0->1->0->0->0->0->0->0-> >0->0	1
37	1	1	1->1->0->1->1->0->0->0->0->0->0->0->0->0->1-> >0->0->0->0->0->0->0->0->0->0->0->0->0->0->0->0-> >0->1->0->0->0	1
41	1	1	1->0->0->0->1->1->1->1->0->1->1->1->1->1->1->0-> >1->1->1->1->1->0->1->1->1->0->1->1->1->1->1->1->1-	1

n stays	Subjects	Unique patterns	Most common sequence	Count
			>1->0->1->1->1->1->1->1->1	